PROJECT CONCEPT NOTE

ISOKOLINK RWANDA

A Digital Platform Connecting Agricultural Producers, Sellers, and Buyers Through Market Information and Price Transparency

Project Name: ISOKOLINK RWANDA
Project Category: Agriculture, Digital Innovation, Market Access, and Food Systems
Project Type: Web and Mobile Application
Target Country: Rwanda
Target Users: Farmers, agricultural cooperatives, produce traders, retailers, restaurants, households, and other food buyers
Technology: React.js, React Native, Node.js, Express.js, TypeScript, and MySQL
Platforms: Website, Android, and iOS
Project Status: Under Development

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1. EXECUTIVE SUMMARY

ISOKOLINK RWANDA is a digital marketplace designed to connect agricultural producers, food sellers, and buyers across Rwanda through an accessible and reliable digital platform.

The project aims to address challenges in agricultural marketing, including difficulties finding buyers, limited access to market information, inadequate price transparency, inefficient communication between sellers and buyers, and the loss of agricultural products when they cannot be sold in time.

In Rwanda, farmers and other agricultural producers work hard to produce food such as tomatoes, Irish potatoes, beans, maize, fruits, vegetables, and other agricultural products. However, producing food does not automatically guarantee that the producer will find a market for it.

During harvest periods, several producers may bring similar products to the market simultaneously. When the available supply exceeds immediate demand, some farmers may struggle to find buyers at an acceptable price. Perishable products, such as tomatoes, leafy vegetables, and ripe fruits, can deteriorate quickly if they remain unsold. In some situations, farmers may be forced to sell their products at very low prices to avoid losing their harvest. Others may experience partial or complete losses because the products cannot be sold or stored appropriately.

At the same time, buyers may struggle to identify sellers offering the products they need at reasonable prices. They may spend considerable time visiting different markets, contacting multiple sellers, or relying on limited information when deciding where to purchase agricultural products.

ISOKOLINK RWANDA seeks to reduce these challenges by providing a digital platform where sellers can publish agricultural product listings, buyers can discover available products and compare listed prices, and both parties can communicate directly.

The platform will include a web application and mobile applications for Android and iOS. Sellers will be able to create accounts, upload product information and images, indicate prices and available quantities, and submit listings for approval. An administrator will review submitted listings before they become publicly visible. Buyers will be able to search for products, compare prices, view seller information, and communicate with sellers through the platform.

By improving access to market information and facilitating connections between producers and buyers, ISOKOLINK RWANDA aims to support more efficient agricultural marketing, reduce avoidable post-harvest losses, improve price transparency, and contribute to stronger agricultural value chains in Rwanda.

The project does not aim to guarantee that every listed product will be sold. Instead, it seeks to make it easier for sellers and buyers to find one another and make better-informed market decisions.

2. BACKGROUND AND CONTEXT

Agriculture plays an important role in Rwanda's economy and food system. Farmers and agricultural businesses produce a wide range of products that supply households, local markets, restaurants, retailers, and other businesses.

Despite the importance of agricultural production, challenges can arise between harvesting a product and successfully selling it to the final buyer.

Agricultural marketing involves more than producing food. Producers must also identify potential buyers, communicate product availability, negotiate prices, arrange transportation where necessary, and deliver products before they deteriorate.

These activities can become difficult when sellers have limited information about market demand or when buyers cannot easily discover available products.

For example, a farmer harvesting tomatoes may have a large quantity ready for sale. If the farmer does not know which traders, restaurants, retailers, or other buyers are looking for

tomatoes, finding customers may take time. If the tomatoes remain unsold for too long, their quality can deteriorate, reducing their market value or making them unsuitable for sale.

Similarly, a farmer harvesting Irish potatoes may discover that many other farmers are selling the same product during the same period. The increased supply in a particular market may place downward pressure on prices. Without sufficient information about alternative buyers or markets, the farmer may have limited options.

These challenges can also affect consumers and businesses. A restaurant may need tomatoes, onions, potatoes, or other ingredients but may not know which sellers currently have sufficient quantities available. A retailer may spend time contacting several suppliers before finding a suitable offer.

Digital technology offers an opportunity to improve the flow of information between these market participants.

ISOKOLINK RWANDA proposes a practical digital solution that allows agricultural sellers to display their products and prices online while enabling buyers to discover available products and contact sellers more conveniently.

By making market information more accessible, the platform aims to help participants make better decisions about where to sell, where to buy, and which available offers best meet their needs.

3. PROBLEM STATEMENT

Agricultural producers and food buyers in Rwanda may face difficulties accessing timely, reliable, and convenient information about available products, potential buyers, and market prices.

One important challenge occurs when farmers successfully produce crops but cannot find sufficient buyers when the products are ready for harvest.

For instance, a farmer may invest money, labour, seeds, fertilizers, and time in growing tomatoes, cabbages, fruits, or other crops. When harvesting begins, the farmer needs to sell the produce quickly to recover production costs and earn income. However, if the farmer cannot identify enough buyers, the products may remain unsold.

This situation becomes particularly serious for perishable agricultural products. Tomatoes, leafy vegetables, ripe fruits, and similar crops can lose freshness and quality when they remain unsold for an extended period. Without suitable storage, transportation, or access to alternative markets, some of these products may spoil before reaching consumers.

Consequently, farmers may be forced to sell their harvest at prices significantly below their expectations, accept offers that provide little profit, or experience losses when part of the harvest becomes unusable. In some cases, products may be discarded even though potential buyers elsewhere could have used them.

These problems do not necessarily arise because there are no buyers for the products. Sometimes, the challenge is that sellers and potential buyers cannot easily find one another at the right time.

For example, while one farmer is struggling to sell a large quantity of Irish potatoes, a retailer, restaurant, trader, or household in another area may be searching for Irish potatoes but may not know that the farmer has them available. Limited connections and information can prevent a mutually beneficial transaction.

Another challenge is the limited visibility of prices and available offers across different sellers and markets. When farmers have insufficient information about what other buyers are offering, they may find it difficult to negotiate a reasonable price or identify a better selling opportunity. Likewise, buyers may pay more than necessary or spend additional time searching for products because they cannot conveniently compare available offers.

Communication can create further difficulties. Sellers may depend on personal contacts, physical market visits, telephone calls, or informal networks to advertise their products. These approaches can be useful, but they may not reach all potential customers, particularly those outside a seller's immediate network.

Small-scale producers and traders may also lack simple digital tools for displaying product images, specifying quantities, communicating prices, and updating availability. As a result, buyers may have difficulty knowing which products are available, their current prices, their quantities, and how to contact the sellers.

These challenges can contribute to inefficient market connections, missed sales opportunities, avoidable food losses, reduced income for some producers, and unnecessary search costs for buyers.

The central problem is the gap between agricultural production and access to markets. Farmers and other sellers need better ways to reach potential buyers, while buyers need more convenient access to information about available agricultural products, prices, quantities, and sellers.

ISOKOLINK RWANDA is proposed as a digital platform to help address this gap by connecting sellers and buyers, improving access to market information, supporting price comparison, and facilitating direct communication.

4. PROJECT JUSTIFICATION

ISOKOLINK RWANDA is justified by the need to improve the way agricultural products are advertised, discovered, and exchanged in local markets.

A digital marketplace can provide several benefits.

First, it can expand the visibility of agricultural products beyond the immediate physical market or personal network of the seller. A farmer who previously relied on a few known traders could publish product information that is accessible to additional potential buyers.

Second, it can make market information easier to access. Buyers can view available listings and compare listed prices instead of depending entirely on physical visits or individual telephone inquiries.

Third, it can support faster communication. Buyers can contact sellers directly to ask about quantities, quality, availability, collection arrangements, and other purchasing conditions.

Fourth, it can help sellers identify additional selling opportunities before perishable products deteriorate. Although listing a product does not guarantee a sale, increasing the number of potential buyers who can discover the product may improve the seller's chances of finding a suitable customer.

Fifth, it can support more informed decisions. Sellers may gain better visibility into competing offers, while buyers may identify products that meet their needs and budgets.

Finally, the project can contribute to agricultural digitalization by providing an accessible online service designed around the practical needs of Rwanda's agricultural market participants.

5. PROJECT VISION

To become a trusted digital platform that connects agricultural producers, food sellers, and buyers across Rwanda, making agricultural markets more accessible, transparent, and efficient.

6. PROJECT MISSION

To use digital technology to connect sellers with potential buyers, improve access to agricultural market information, support price comparison, and reduce avoidable losses associated with unsold agricultural products.

7. PROJECT OBJECTIVES

7.1 General Objective

To develop a secure and accessible digital marketplace that improves connections between agricultural sellers and buyers in Rwanda by providing convenient access to product listings, prices, quantities, seller information, and communication tools.

7.2 Specific Objectives

The project aims to:

1. Develop a digital platform through which farmers, cooperatives, traders, and other sellers can advertise available agricultural products.
2. Enable buyers to search for agricultural products and discover available offers more conveniently.
3. Support price transparency by allowing users to compare listed prices and product details across participating sellers.
4. Improve communication between agricultural sellers and potential buyers through integrated messaging and seller contact options.
5. Help sellers reach a wider pool of potential customers, including traders, retailers, restaurants, institutions, and households.
6. Contribute to reducing avoidable post-harvest losses by improving product visibility and helping sellers identify potential buyers before products deteriorate.
7. Provide an administrator approval process to improve the quality, reliability, and appropriateness of public product listings.
8. Develop web, Android, and iOS applications that allow users to access the platform through different devices.
9. Establish a foundation for collecting and analysing marketplace information to support future improvements in agricultural marketing.
10. Promote more informed purchasing and selling decisions by making product availability, listed prices, quantities, and seller information easier to access.

8. PROPOSED SOLUTION

ISOKOLINK RWANDA will provide a digital marketplace where agricultural sellers can publish product listings and buyers can discover and compare available offers.

The platform will consist of three main components:

8.1 Web Application

The website will allow users to access the marketplace through a web browser on a computer, tablet, or smartphone.

Users will be able to register, log in, browse products, search for offers, view product details, communicate with sellers, and manage their accounts according to their permissions.

8.2 Android Mobile Application

The Android application will provide mobile access to the platform. Users will be able to search for agricultural products, view prices, contact sellers, manage listings where authorized, and receive supported notifications.

8.3 iOS Mobile Application

The iOS application will provide equivalent core marketplace functionality for iPhone users.

Both mobile applications will connect to the same backend and database as the website, allowing users to access the same accounts, product listings, and conversations across supported platforms.

8.4 How the Solution Will Work

The platform will follow a straightforward process:

1. A user creates an account as a buyer or seller.
2. A seller submits an agricultural product listing containing the product name, category, description, images, price, measurement unit, available quantity, and relevant location information.
3. The listing is sent to the administrator for review.
4. The administrator approves or rejects the listing. Rejected listings can include an explanation to help the seller understand what needs to be corrected.
5. Once approved, the listing becomes visible to buyers.
6. Buyers search for products, review available offers, and compare prices and other relevant details.
7. An interested buyer contacts the seller through the platform's messaging system or an available contact option.

8. The buyer and seller discuss the product, availability, price, delivery or collection arrangements, and other transaction details.
9. The parties agree on how to complete the transaction.
10. Sellers can update product availability so that information remains as accurate as possible.

The initial platform will primarily facilitate discovery, comparison, and communication. Unless additional functionality is implemented, it will not itself guarantee product delivery, payment, or completion of a transaction.

9. TARGET USERS AND BENEFICIARIES

9.1 Farmers and Agricultural Producers

Farmers will be able to advertise their harvest, display available quantities, communicate prices, and reach potential buyers beyond their immediate networks.

The platform may be particularly useful to producers who struggle to find buyers during harvest periods.

9.2 Agricultural Cooperatives

Cooperatives will be able to publish information about products available from their members and connect with potential wholesale buyers.

This may help cooperatives communicate available quantities and identify new commercial relationships.

9.3 Agricultural Traders and Wholesalers

Traders and wholesalers will be able to discover available products, compare offers, contact suppliers, and identify potential sources of agricultural produce.

9.4 Retailers and Market Vendors

Retailers will be able to search for suppliers offering products that they need for resale.

9.5 Restaurants, Hotels, and Food Businesses

Restaurants, hotels, and other food businesses will be able to identify potential suppliers and compare listed prices before contacting sellers.

9.6 Households and Individual Consumers

Households will be able to discover available agricultural products and contact sellers directly, subject to the quantities and selling arrangements offered.

9.7 Administrators

Administrators will oversee the platform, review submitted listings, manage user-related issues within their permissions, and help maintain the quality of marketplace information.

10. KEY FEATURES OF THE PLATFORM

10.1 User Registration and Authentication

Users will be able to create accounts and securely log in.

The platform will support appropriate authentication, account verification, password recovery, and account management.

A user will have one account. A buyer who later decides to sell agricultural products should be able to activate seller functionality without creating a duplicate account.

Access to features will depend on the user's permissions.

10.2 Product Listing Management

Sellers will be able to submit agricultural products with relevant details, including:

- Product name.
- Product category.
- Description.
- Product images.
- Price.
- Measurement unit.
- Available quantity.
- Seller information.
- Location, where appropriate.
- Listing status.

Sellers should be able to update product information and availability according to the platform's rules.

10.3 Administrator Approval

New listings will be reviewed before publication.

The administrator will be able to approve or reject listings and provide reasons when necessary.

This process is intended to reduce inappropriate, misleading, or incomplete listings.

Administrator permissions will be enforced by the backend rather than relying only on what is displayed in the application.

10.4 Product Search and Filtering

Buyers will be able to search for products using product names and categories.

Where data is available, additional filters may include:

- Price range.
- Location.
- Availability.
- Product category.
- Seller type.
- Measurement unit.

These features will make it easier to find relevant products.

10.5 Price Comparison

The platform will display prices submitted by sellers so that buyers can compare available offers.

For example, a buyer searching for Irish potatoes may find several approved listings from different sellers and compare their prices, quantities, locations, and other product details.

To make comparisons meaningful, the platform should display the measurement unit clearly and avoid comparing prices for different quantities as though they were equivalent.

Listed prices will represent the information supplied by sellers and should not automatically be presented as official market prices.

10.6 Direct Communication

The platform will provide messaging between buyers and sellers.

Buyers will be able to ask about product quality, available quantities, current prices, delivery arrangements, and other relevant conditions.

Sellers will be able to respond to inquiries and discuss possible transactions.

Authorized administrators will be able to communicate with users and manage reported issues.

Access to conversations will be controlled according to user permissions.

10.7 Notifications

Where supported, users may receive notifications about new messages, listing approvals, listing rejections, and important account events.

Notifications will help users respond to relevant activities more promptly.

10.8 User Profiles

Users will have profiles containing appropriate account and contact information.

Seller profiles may include business information, location, and other relevant details.

The platform will protect private account information and display only information that is appropriate to share publicly.

10.9 Availability Management

Sellers will be able to update product quantities and indicate when products are no longer available.

This will help reduce situations in which buyers contact sellers about products that have already been sold.

10.10 Future Analytics

As the platform grows, aggregated marketplace information may support insights into:

- Commonly listed products.
- Differences between listed offers.
- Product availability across locations.
- Changes in listing prices over time.
- Categories with high levels of marketplace activity.

Such insights will depend on the quantity, quality, and consistency of the data collected.

11. TECHNOLOGY AND SYSTEM ARCHITECTURE

ISOKOLINK RWANDA will use a technology architecture that supports web access, mobile applications, centralized data management, and future expansion.

11.1 Frontend Technologies

The existing website uses:

- HTML5.
- CSS3.
- JavaScript.
- TypeScript.
- React.js.
- Vite.
- Tailwind CSS.
- React Router.
- Axios.
- React Hook Form and Zod, where applicable.

These technologies support the web-based user interface.

11.2 Mobile Application Technologies

The mobile application will use:

- React Native.
- Expo.
- TypeScript.
- Expo Router or React Navigation.
- Axios.
- Expo SecureStore.
- Appropriate mobile form validation and image-upload tools.

React Native and Expo will support development for Android and iOS from a shared mobile codebase.

11.3 Backend Technologies

The backend uses or is designed around:

- Node.js.
- Express.js.
- TypeScript.
- REST APIs.
- Server-side validation.
- Authentication and authorization.
- MySQL database integration.

The backend will process requests, enforce business rules, manage users and listings, control approval workflows, and provide data to the website and mobile applications.

11.4 Database

MySQL will provide centralized storage for information such as:

- User accounts.
- Roles and permissions.
- Product listings.
- Product categories.
- Product images or image references.
- Listing approval records.
- Conversations and messages.
- Relevant user preferences.
- Other required marketplace records.

The exact schema will follow the existing implementation and any carefully reviewed migrations.

The mobile applications will not connect directly to MySQL. All data access will go through the backend API.

11.5 Shared Platform Architecture

The website, Android application, and iOS application will use the same backend and database.

This means that a product approved through the administrative system can become available on supported platforms without requiring separate product records for each application.

Likewise, users should be able to use the same accounts and access their relevant conversations across supported devices.

11.6 Security

The system will use appropriate security measures, including secure authentication, role-based authorization, validated inputs, protected API endpoints, secure handling of credentials, and HTTPS in production.

Private configuration values will be stored in protected environment variables on the server rather than exposed in frontend or mobile application code.

12. EXPECTED PROJECT OUTPUTS

The main outputs will include:

1. A functional ISOKOLINK RWANDA website.
2. An Android-compatible mobile application.
3. An iOS-compatible mobile application.
4. A centralized backend API.
5. A MySQL database supporting marketplace operations.
6. Secure user registration and authentication.
7. A product listing and management system.
8. An administrator approval workflow.
9. Product search and price-comparison functionality.
10. Buyer-seller messaging.
11. User profiles and availability management.
12. Project documentation and installation instructions.
13. Testing procedures and deployment guidance.

The final functionality will depend on successful implementation, testing, and configuration of the relevant services.

13. EXPECTED OUTCOMES AND IMPACT

13.1 Improved Market Access

Sellers will have an additional channel through which to advertise their agricultural products and reach potential buyers.

13.2 Reduced Risk of Avoidable Product Losses

By making product availability visible to more potential buyers, the platform may help some sellers identify customers before perishable products deteriorate.

The actual reduction in losses will depend on adoption, buyer demand, storage conditions, transportation, and the speed at which transactions take place.

13.3 Better Access to Price Information

Buyers will be able to compare prices and other details across available listings.

Sellers may also gain greater visibility into competing offers, helping them make more informed selling decisions.

13.4 Improved Communication

Direct messaging will help reduce the effort required to identify and contact potential trading partners.

13.5 More Informed Purchasing Decisions

Buyers will be able to review available products, listed prices, quantities, and seller details before deciding whom to contact.

13.6 Expanded Business Opportunities

Farmers, cooperatives, and traders may discover new customers and commercial relationships through the platform.

13.7 Contribution to Agricultural Digitalization

The project will encourage the use of digital tools in agricultural marketing and provide a foundation for future digital agricultural services.

14. IMPLEMENTATION APPROACH

The project will be implemented in phases.

Phase 1: Requirements Analysis and System Review

Activities:

- Review the existing website and backend.
- Review the database schema.
- Identify the main user roles.
- Confirm existing API endpoints.
- Identify missing functionality.
- Finalize requirements and technical architecture.

Expected output: A documented implementation plan and confirmed technical requirements.

Phase 2: Backend and Database Preparation

Activities:

- Review and improve authentication where necessary.
- Implement missing API endpoints.
- Confirm product submission and approval workflows.
- Implement or improve messaging functionality.
- Add required database migrations safely.
- Improve validation, authorization, and error handling.

Expected output: A secure and functional backend supporting the required marketplace features.

Phase 3: Website Improvements

Activities:

- Implement or refine marketplace pages.
- Improve product search and filtering.
- Complete seller listing management.
- Improve the administrator dashboard.
- Ensure the website uses live backend data.

Expected output: A functional web marketplace integrated with the backend.

Phase 4: Mobile Application Development

Activities:

- Create the React Native and Expo application.
- Develop onboarding and authentication screens.
- Build buyer and seller dashboards.
- Implement product browsing and price comparison.
- Integrate listing submission and approval status.
- Implement messaging and profiles.
- Configure API connectivity.
- Prepare Android and iOS compatibility.

Expected output: A mobile application that can be tested on Android and prepared for iOS builds.

Phase 5: Testing and Quality Assurance

Activities:

- Test user registration and login.
- Test product submissions and approval.
- Test product searches and price comparison.
- Test messaging and account management.
- Test authorization and security.
- Test different screen sizes.
- Test Android and iOS compatibility.
- Resolve identified defects.

Expected output: Tested applications with documented results and remaining limitations.

Phase 6: Deployment and Launch Preparation

Activities:

- Configure production hosting and HTTPS.
- Deploy the backend and database.
- Publish the website.
- Prepare Android application builds.
- Prepare iOS builds using an appropriate Apple-supported build and signing environment.
- Prepare application documentation.
- Train initial users where possible.

Expected output: A deployment-ready platform and mobile application builds.

Phase 7: Monitoring and Continuous Improvement

Activities:

- Monitor performance and reliability.
- Collect user feedback.
- Improve usability.
- Address security issues.
- Refine product search and market information features.
- Plan future enhancements based on evidence and user needs.

Expected output: A sustainable improvement process based on real usage and feedback.

15. PROJECT SCOPE

15.1 Included in the Initial Scope

The initial project will focus on:

- User registration and authentication.
- Buyer and seller accounts.
- A shared user account when a buyer becomes a seller.
- Agricultural product listings.
- Product images and descriptions.

- Administrator approval of listings.
- Product search and filtering.
- Price comparison.
- Seller contact information.
- Buyer-seller messaging.
- User profiles.
- Product availability updates.
- Web, Android, and iOS access.
- Basic administration and monitoring.

15.2 Potential Future Enhancements

The following features may be considered after the core platform is functional:

- In-app payments.
- Delivery and logistics coordination.
- Verified seller profiles.
- Ratings and reviews.
- Buyer order management.
- Digital receipts.
- Market price trend analysis.
- Demand forecasting.
- Automated recommendations.
- Notifications about relevant product listings.
- Integration with cooperatives and agricultural organizations.
- Support for additional local languages.
- Advanced tools for reducing food waste.
- Integration with relevant agricultural information services.

These features will require separate planning, validation, and implementation.

16. PROJECT RISKS AND MITIGATION MEASURES

16.1 Limited User Adoption

Risk: Some farmers and buyers may be unfamiliar with digital marketplaces or may prefer existing trading methods.

Mitigation: Provide a simple interface, practical onboarding, user education, and support for mobile devices.

16.2 Limited Internet Access

Risk: Some potential users may experience unreliable internet connectivity.

Mitigation: Optimize the application for mobile networks, reduce unnecessary data usage, and provide clear loading and retry behaviour.

16.3 Inaccurate Product Information

Risk: Sellers may submit incorrect prices, quantities, or product descriptions.

Mitigation: Use listing validation, administrator review, seller update tools, reporting mechanisms, and appropriate account controls.

16.4 Products Becoming Unavailable

Risk: A product may be sold before the seller updates its listing.

Mitigation: Allow sellers to update availability easily and provide clear information about when listings were last updated.

16.5 Continued Post-Harvest Losses

Risk: Connecting sellers with buyers may not be sufficient when demand is low or storage and transportation are inadequate.

Mitigation: Present the platform as a market-access tool rather than a complete solution to every cause of food loss. Explore future partnerships and logistics support where feasible.

16.6 Security and Privacy Risks

Risk: Unauthorized access or misuse of personal information could undermine user trust.

Mitigation: Implement secure authentication, server-side authorization, input validation, appropriate access controls, and secure data handling.

16.7 Price Differences and Misunderstandings

Risk: Users may misunderstand price comparisons when sellers use different measurement units, product grades, or selling conditions.

Mitigation: Require clear measurement units and display relevant product details, quantities, and conditions alongside prices.

16.8 Technical and Operational Costs

Risk: Hosting, maintenance, storage, notifications, and ongoing development may create costs.

Mitigation: Use an appropriate initial infrastructure, monitor resource usage, document the system, and plan for long-term maintenance.

17. MONITORING AND EVALUATION

The project's performance should be evaluated using measurable indicators.

Potential indicators include:

- Number of registered users.
- Number of active sellers.
- Number of active buyers.
- Number of agricultural products submitted.
- Number and percentage of listings approved.
- Number of active product listings.
- Number of buyer-seller conversations.
- Number of reported completed transactions, where reliable reporting is available.
- Time taken to review submitted listings.
- Number of users returning to the platform.
- User satisfaction.
- Number of sellers reporting that the platform helped them find potential buyers.
- User-reported experiences relating to unsold products and post-harvest losses.

Where feasible, the project may collect information about whether sellers found buyers more quickly or reduced losses after using the platform.

These outcomes should be measured through user feedback and appropriate data collection rather than assumed in advance.

Initial targets should be established after the project team assesses available resources, the target market, and the results of early testing.

18. SUSTAINABILITY AND BUSINESS MODEL

For long-term sustainability, ISOKOLINK RWANDA may consider a business model that allows the platform to remain accessible while covering operational and development costs.

Potential revenue sources include:

18.1 Premium Business Services

Advanced tools for cooperatives, wholesalers, or businesses with large product catalogues could be offered through paid plans.

18.2 Promoted Listings

Sellers may be offered clearly labelled promotional placements that improve product visibility, subject to fair marketplace policies.

18.3 Business Partnerships

The platform may establish partnerships with agricultural cooperatives, retailers, restaurants, logistics providers, and relevant institutions.

18.4 Optional Value-Added Services

Future services could include business analytics, advanced inventory tools, and market information services.

The initial business model should be tested with users before fees are introduced. Any revenue model should avoid unnecessarily excluding small-scale farmers from basic marketplace access.

19. STAKEHOLDERS AND PARTNERS

Potential stakeholders include:

- Individual farmers.
- Agricultural cooperatives.
- Produce traders and wholesalers.

- Retailers and market vendors.
- Restaurants and food businesses.
- Individual consumers.
- Technology and digital innovation organizations.
- Agricultural development organizations.
- Relevant public institutions.
- Potential investors and development partners.
- Delivery and logistics service providers.

Partnerships may help the project reach more users, improve adoption, and address challenges that a digital platform alone cannot solve.

Specific partnerships will need to be identified and confirmed as the project develops.

20. ETHICAL, SOCIAL, AND ENVIRONMENTAL CONSIDERATIONS

ISOKOLINK RWANDA aims to improve access to agricultural market information without unfairly excluding users with limited technical skills or resources.

The platform should:

- Use a simple and accessible interface.
- Protect users' personal information.
- Provide clear information about prices and measurement units.
- Avoid making unsupported claims about market prices.
- Apply consistent listing approval rules.
- Provide appropriate reporting mechanisms.
- Avoid guaranteeing sales or profits.
- Avoid claiming that the platform alone will eliminate food waste.
- Encourage accurate product descriptions and availability updates.
- Consider the needs of small-scale producers and users with limited internet access.

By helping sellers and buyers identify one another, the platform may contribute to more efficient use of agricultural products. However, environmental benefits should be assessed using evidence from actual use.

21. EXPECTED LONG-TERM IMPACT

In the long term, ISOKOLINK RWANDA aims to contribute to a more connected agricultural marketplace in which producers and buyers can access relevant information and communicate more efficiently.

The platform has the potential to help farmers expand their customer networks, support more informed price comparisons, and reduce some losses associated with difficulties finding buyers.

It may also help retailers, restaurants, and households identify available agricultural products more conveniently.

As participation grows, the platform could provide a stronger foundation for digital agricultural commerce and future services related to market intelligence, logistics, and agricultural supply chains.

Achieving these benefits will depend on adoption, the reliability of listings, buyer demand, internet accessibility, and effective implementation.

22. CONCLUSION

ISOKOLINK RWANDA is a digital innovation project designed to address the gap between agricultural production and access to markets in Rwanda.

Farmers and other agricultural producers can invest substantial effort and resources in producing food, yet still experience difficulties finding buyers when their products are ready for sale. When these difficulties persist, perishable crops may deteriorate, farmers may accept unfavourable prices, and valuable agricultural products may go to waste.

At the same time, buyers need convenient ways to discover available products, compare prices, identify suppliers, and communicate with sellers.

ISOKOLINK RWANDA seeks to address these challenges by providing a digital marketplace that connects agricultural sellers and buyers through product listings, price comparison, direct communication, and an administrator approval process.

Through a shared web, Android, and iOS platform, the project aims to make agricultural market information more accessible and create additional opportunities for sellers and buyers to connect.

The project recognizes that digital technology cannot independently solve every agricultural marketing challenge. However, by improving the visibility of available products and facilitating communication between market participants, ISOKOLINK RWANDA can provide a practical foundation for better-informed trading decisions and more efficient agricultural marketing.

ISOKOLINK RWANDA — Connecting Agricultural Sellers and Buyers for Better Market Access, Price Transparency, and Reduced Food Loss.